

FINAL EXAM

ECO403/503–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

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Instructions & Rules: The exam has four problems, each worth 25 points, and a bonus question, worth 5 points. You need to write your solutions in the provided answers booklet. You are free to use as much scrap paper as you need to sketch and draft your solutions, but only what you write in the answers booklet will be considered. Write your final solutions, neatly and succinctly. Aim for rigorousness as well as efficiency. Any answer that is not supported by an argument will not be valid. Your solutions will be graded on the basis of accuracy, correctness, coherency, conciseness, as well as exposition.

As discussed previously, the rules for the exam are:

- Consult your class notes and slides printouts.
- Consult any book that I provided as reference.
- Consult any other book/reference that you deem suitable, either printed or online.
- Consult with your ECON403/503 classmates. You are not permitted to consult with people outside the class.
- Any reference that you use, should be acknowledged and cited where applicable. For example, if you consult a webpage or a book, you should give credit and put the proper citation. If you get an idea from a classmate, you should acknowledge it.
- You should deliver the answers booklet in person to me before 5 PM. I will be in the office at 4 PM to start receiving the exams. If you want to deliver earlier than that, let me know.

DO NOT OPEN THIS BOOKLET UNTIL TOLD!

Here is a picture of my cat, Panqueque, to give you good luck!



Bonus question

Write, in English, your favorite verse of your favorite song. *Write it in the space provided in the answer booklet.*

Problem 1: A procurement contest.

Consider a research and development race for a government procurement contract for defense weapons. In this real-life scenario, different companies make costly and irreversible investments to produce prototypes, which the government then evaluates and chooses the winner. The government then shares the prototype's production cost with the winner only. In a case like this, the winner of the research and development contest faces a lower marginal cost than the loser.

To model the above situation, consider the following *complete information* game with two players. The players are denoted by i and j . Both players value winning the R&D race as W . Players simultaneously expend irreversible and costly efforts $x_i \geq 0$ and $x_j \geq 0$. The probability of player i winning the contest is given by

$$p_i(x_i, x_j) = \begin{cases} \frac{x_i}{x_i + x_j} & \text{if } x_i > 0, x_j > 0 \\ 1/2 & \text{if } x_i = x_j = 0. \end{cases}$$

Contingent upon winning or losing, the payoff for player i is given by:

$$u_i(x_i, x_j) = \begin{cases} W - c_1 x_i & \text{with probability } p_i(x_i, x_j), \\ -c_2 x_i & \text{with probability } 1 - p_i(x_i, x_j), \end{cases}$$

where $W > 0$ and the cost parameters (c_1, c_2) satisfy $0 < c_1 < c_2$. All this information is *common knowledge* to the players.

1. Compute the expected payoff $U_i(x_i, x_j) = E[u_i(x_i, x_j)]$ of player i , when x_i and x_j are strictly positive.
2. Assume that the two players are exerting strictly positive efforts. Compute the best response function of both players.
3. Using the best responses, solve for the equilibrium effort levels.
4. How do the equilibrium effort levels depend on the parameters c_1 and c_2 ? Interpret this comparative statics result in terms of the story this model is trying to explain.

Problem 2: A discrete auction where the seller can restrict the bids.

Consider a seller that can sell an object to one of two bidders. The private values of the two buyers are independent and uniformly distributed over the set $\{0, 1\}$. The seller will organize a first-price auction. The novelty is that now the seller can constrain the bids to be only 0 or x , where $x \in (0, 1]$ is a parameter chosen by the seller. Ties are resolved with a uniform lottery. After the seller publicly announces the value of x , he runs the auction.

1. Represent the first-price auction that the buyers participate in as *Bayesian game*, i.e., represent it as a collection of matrices. Be sure to also include a description of what the players know and their beliefs.
2. What is a pure-strategy in this Bayesian game? What is a mixed-strategy in this game?

For every value of $x \in (0, 1]$, the goal is to find *all* the Bayesian equilibria in the first-price auction above. Note that since the players are symmetric, all equilibrium profiles will be symmetric. Thus, a hint is that anything you do for one player, applies to the other.

You will start with pure-strategy equilibria.

3. For what values of x is the strategy “bid 0 when value 0, bid x when value 1” a strictly dominant strategy?
4. For what values of x is the strategy: “bid 0 always” a strictly dominant strategy?
5. For what values of x are there multiple equilibria and what form they take?

Now you will look for a mixed-strategy equilibrium.

6. For what values of x is there a mixed-strategy equilibrium and what is it?

You will explore now some expected revenue considerations for the seller, who can choose the parameter $x \in (0, 1]$. You will need to look at how expected revenue changes as the value of x moves.

7. Suppose that the seller, when there are multiple equilibria, can lead the players to his *preferred equilibrium*. What is the expected revenue maximizing value of x ?
8. Now suppose that, whenever there are multiple equilibria, the players will choose to play the equilibrium that is the *worst for the seller*. What is now the expected revenue maximizing value of x ?

Problem 3: A Private Values Auction with a non-independent distribution of values.

Consider a private-value auction with two bidders. However, in contrast with the independent distribution of values case from the SIPV model, suppose now that the distribution of values is not independent, but follows the following prior:

$$f(v_1, v_2) = \frac{1}{2} I_{A_1 \cup A_2}(v_1, v_2),$$

where $A_1 = \{(v_1, v_2) \in [0, 1]^2\}$ and $A_2 = \{(v_1, v_2) \in [1, 2]^2\}$, and $I_{A_1 \cup A_2}(v_1, v_2)$ is the indicator function of the set $A_1 \cup A_2$.

1. Give the complete *belief space* description of the situation described above. Be sure to specify what are the signalling functions of the bidders.
2. Compute bidder i 's posterior beliefs after they receive their signal $V_i = v_i$: show that if $v_i \in [0, 1)$ the posterior beliefs are uniformly distributed over $[0, 1]$; and that if $v_i \in [1, 2]$ the posterior beliefs are uniformly distributed over $[1, 2]$.
3. Give a (semi) formal argument that if $(v_1, v_2) \in A_1$, it is common knowledge between the bidders that their values are in A_1 and that their posterior beliefs about the value of their respective opponents are identically distributed as independent uniform random variables over $[0, 1]$. Conclude something similar for bidders whose values are $(v_1, v_2) \in A_2$: it is common knowledge that their values are in A_2 and that their posterior beliefs about the value of their respective opponents are identically distributed as independent uniform random variables over $[1, 2]$. *Hint: you can use the results from items 8 and 9 from the Beliefs Problem List.*
4. Consider a first-price auction. Find the symmetric, monotonically increasing, right-continuous, equilibrium bidding strategy. *Hint: Use 2 and 3 above to simplify the problem to two cases that can be analyzed with the tools from the SIPV model. The equilibrium bidding function will be continuous, except at a point of discontinuity at 1 in which it will jump upwards.*
5. Compute the expected revenue for the first-price auction at the symmetric equilibrium you found in item 4.
6. Consider a second-price auction, and consider its symmetric equilibrium in weakly-dominant strategies. Compute the expected revenue for the second-price auction. Are the expected revenues for the FPA and SPA the same for this exercise?

Problem 4: A common-value auction.

Consider the general auction model from definition 5.1¹. Suppose that there are only two buyers. Furthermore, suppose that each buyer assigns a *common value* to the object being sold. Such a value is given by

$$V = \frac{T_1 + T_2}{2},$$

where T_1 and T_2 are two independent and identically distributed uniform random variables over the interval $[0, 1]$. The realization of the random variables T_1 and T_2 is private information to each respective buyers. Thus, this is a model with *interdependent values*, and in this problem you will explore the consequences of this extension.

1. Begin by describing explicitly the belief space component of the Bayesian Auction. Be very clear about what is the state space, the signalling functions and the prior over the states. *Hint: note that conditional on the realization of T_1 and T_2 , the value V is no longer random. Thus, the prior is quite simple, since it is a product of independent and identically distributed random variables.*

Now consider a first-price auction. You will now look for a symmetric equilibrium. Towards this end, suppose that $\beta_I(v)$ is a continuous, monotonically increasing, and differentiable bidding function for the FPA.

2. Suppose that player 2 follows the strategy β_I . Calculate an expression for the expectation of V conditional on the realization of $T_1 = t_1$ and on the event that player i wins the auction, i.e., compute $E[V|T_1 = t_1, \text{Player } i \text{ wins the auction.}]$
3. Use the expression in item 2 to compute the *interim expected payoff* of player 1.
4. Use the expression in item 3 to give an heuristic derivation of the symmetric equilibrium strategy in the FPA. *Hint: use the fact that there are only two bidders and you are working with uniform distributions. Do not solve for the general case.*
5. Given what you found in item 4, show that it is indeed a best response for player 1 to follow β_I if player 2 is also following β_I . Use this to conclude that β_I is indeed a symmetric equilibrium.
6. Now consider the second-price auction. Show that $\beta_{II}(t_i) = t_i$, i.e., bidding your type, is a symmetric equilibrium. *Hint: Just show that it is indeed an equilibrium, I am not asking the stronger statement about it being weakly dominant.*
7. Compute the expected revenues at the symmetric equilibrium in both the FPA and the SPA. Does revenue equivalence hold?
8. Consider the following passage²:

In A.D. 193, having killed the Emperor Pertinax, in a bold move the Prætorian Guard proceeded to sell off the entire Roman Empire by means of an auction. The winning bid was a promise of 25,000 sesterces per man to the Guard. The winner, Didius Julianus, was duly declared emperor but lasted for only two months before suffering from what is perhaps the earliest and most extreme instance of the “winner’s curse”: He was beheaded.

In light of the paragraph above, why did we need to condition on the event of *winning* when computing the conditional expected value of V in item 2?

¹Lecture 5, pages 4–5.

²From Krishna, V. (2009). Auction theory. Academic press, p.1